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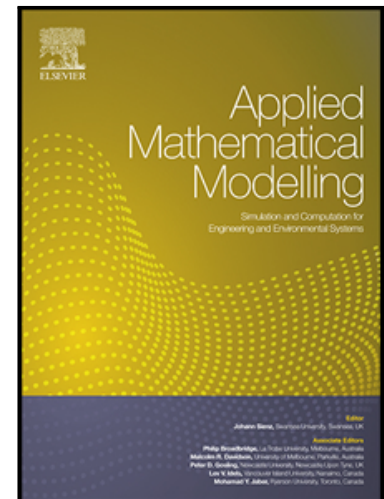
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Highlights

- New nonlocal model for steady-state thermoelastic analysis of FGMs based on peridynamic differential operator
- Declare effect of material gradients and loads on responses of FGM plates by introducing Mori-Tanaka method
- Investigate influences of the degree of nonlocality on stress singularity at crack tip in FGM plates

Nonlocal steady-state thermoelastic analysis of functionally graded materials by using peridynamic differential operator

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Abstract:

In this study, a nonlocal model is presented for the steady-state thermoelastic analysis of functionally graded materials (FGM) by using peridynamic differential operator. The displacement–temperature equations of coupled thermo-elasticity and boundary conditions for a two-dimensional FGM square plate under mechanical and thermal loads are converted from classical local differential forms into nonlocal integral forms with peridynamic differential operator. The temperature, displacement and stress fields are solved by introducing Lagrange multiplier and employing variational analysis. A comparison study is conducted to validate the accuracy and convergence of this nonlocal model by comparing the nonlocal analysis results with finite element results as well as analytical solutions in literature. The effects of different material gradients and loads on the physical fields of FGM plates are investigated further by introducing the Mori-Tanaka method to estimate the effective properties, and the influences of the degree of nonlocality on the stress singularity at the crack tip in FGM plates are analysed finally.

Keywords: Functionally graded materials; peridynamic differential operator; thermo-elasticity; steady-state; stress singularity

1. Introduction

The macroscopic properties of functionally graded materials (FGM) may vary continuously and smoothly in a particular direction, due to the continuous gradient changes of microscopic elements in the same direction, and shows advantages than common materials and composites [1-2]. FGM were initially designed as heat resisting high-performance materials for aerospace structures and fusion reactors, which are able to withstand the ultra-high temperature and alleviate thermal stress, and have shown promising application potential in reactor vessels, aircrafts, space vehicles, and other engineering fields with excellent thermal performances. In this context, it is of significance to investigate the thermo-elasticity problem of FGM structures under mechanical and thermal loads.

Studies on the thermo-elasticity problem of FGMs have been concerned by many scholars since the advent of FGMs. Noda summarized the study on the thermal stresses of FGM structure [3]. Subsequently, he cooperated with other scholars to conduct a lot of researches on the thermo-elasticity problem of FGM structures, including steady-state and transient thermal stress analysis [4-7], thermal bending and buckling [8-10], thermal fracture [11-13]. In recent years, some significant research involving the thermo-elasticity problem of FGM structures are worthwhile to mention. Alibeigloo [14] carried out three-dimensional thermo-elastic analysis of FGM rectangular plates with simply supported edges subjected to thermo-mechanical based on the Fourier series and state-space method. Geometrically nonlinear analysis of FGM circular plates subjected to mechanical and thermal loads was studied by Gunes

and Reddy [15] using finite element method (FEM). Trinh et al. [16] investigated analytical closed-form solutions for thermos-mechanical stability and explicit expressions for free and forced vibration of thin FGM sandwich shells with double curvature resting on elastic bases by using the classical shell theory. A formulation for buckling and geometrically nonlinear analysis of FGM plates under thermo-mechanical loading was presented by Moita et al. [17] implementing the FEM associated with the higher order shear deformation theory.

Due to the heterogeneity of FGM, the solutions of the traditional local theoretical model often have large errors compared with the experimental results, but the non-local model can effectively improve the calculation accuracy [18,19]. Peridynamics (PD), referred to as a reformulation of solid mechanics for analysing both continuous and discontinuous problems based on spatial nonlocal integration [20,21], has been successfully applied to solve the static and dynamic problems as well as crack propagation in various materials and structures at different scales in the past decades [22-29]. It is worth mentioning that Ren et al. [30,31] presented a dual-horizon PD formulation that allows variable horizons and inhomogeneous discretization and completely solves the ‘ghost force’ issue. In particular, PD also has a good performance in fracture of FGM, heat conduction and thermal coupling problems. Cheng et al. [32] analyzed the dynamic fracture failure of FGM with two pre-cracks by using a bond-based PD method. By introducing a new response function, Bobaru et al. [33,34] and Kilic et al. [35] used the bond-based PD theory to establish the transient thermal conductivity formula and solve discontinuous thermal

coupling problem, respectively. Oterkus et al. [36,37] proposed a transient heat conduction method and a fully coupled thermodynamic model in dimensionless form based on the state-based PD theory, and gave the response functions of different dimensional models. Song et al. [38] developed a bond-based PD de-icing model to simulate the coupled thermomechanical ice removal process on frozen structures.

With the further development of PD, many nonlocal differential operator methods [39-45] were presented. As one of the most important representatives in these methods, the peridynamic differential operator (PDDO) was proposed by Madenci [44,45] based on the concept of PD non-local interaction, and has shown unique advantages in handling various problems in the recent 4 years. For typical examples, Dorduncu [46,47] used PDDO and refined zigzag theory to carry out the stress analysis of laminated composite beams and sandwich plates with functionally graded cores. The nonlocal models of low Reynolds number laminar fluid motion and fluid flow coupled with heat transfer were established by Gao et al. [48,49] based on PDDO. Shojaei et al. [50] coupled generalized finite difference method with PDDO and successfully applied it to two-dimensional Helmholtz type problems. The local derivatives can be expediently converted into their nonlocal integral forms by using PDDO theory. Motivated by the research focuses as well as requirements from both FGMs and PDDO, we attempt to employ PDDO to analyse the nonlocal steady-state thermo-elasticity of functionally graded materials in this work.

This paper presents a nonlocal model for the steady-state thermoelastic analysis of FGMs by using peridynamic differential operator. We take a two-dimensional

functionally gradient square plate under mechanical and thermal loads as an example and reformulate the displacement–temperature equations of coupled thermo-elasticity, Hooke’s law for isotropic thermoelastic materials and boundary conditions from their local differential forms into their nonlocal integral forms. By introducing the Lagrange multiplier, employing variational analysis, and using the Intel direct solver, we can obtain the temperature, displacement and stress fields. We compare the presented results with finite element solutions and analytical solutions in the previously published literature to verify the accuracy and convergence of this model. Finally, the Mori-Tanaka method is introduced to estimate the effective properties, and we analyse the effects of different material gradients and loads on the physical fields of the FGM plate. Finally we further investigated the effect of the degree of nonlocality on the stress singularity at the crack tip in FGM plates.

2. Peridynamic differential operator

A scalar field function, $f=f(\mathbf{x})$, is considered in a M -dimensional space. The Taylor series expansion (TSE) of N order of this function, $f(\mathbf{x}')$, at a point, $\mathbf{x}$, without the remainder which is negligibly small can be expressed as [44,45]

$$f(\mathbf{x}') = \sum_{n_1=0}^N \sum_{n_2=0}^{N-n_1} \cdots \sum_{n_M=0}^{N-n_1 \cdots n_{M-1}} \frac{1}{n_1! n_2! \cdots n_M!} \xi_1^{n_1} \xi_2^{n_2} \cdots \xi_M^{n_M} \frac{\partial^{n_1+n_2+\cdots+n_M} f(\mathbf{x})}{\partial x_1^{n_1} \partial x_2^{n_2} \cdots \partial x_M^{n_M}} \quad (1)$$

in which $\mathbf{x}'=\mathbf{x}+\boldsymbol{\xi}$ and $\mathbf{x}' \in H_{\mathbf{x}}$, as shown in Fig. 1 in the case of 3-dimensional space. $H_{\mathbf{x}}$ is the domain of interaction in the material point $\mathbf{x}$, as well as called the family of which the size is established by a characteristic length δ , referred to as the “horizon”, in the Peridynamic theory [20,21] introduced by Silling.

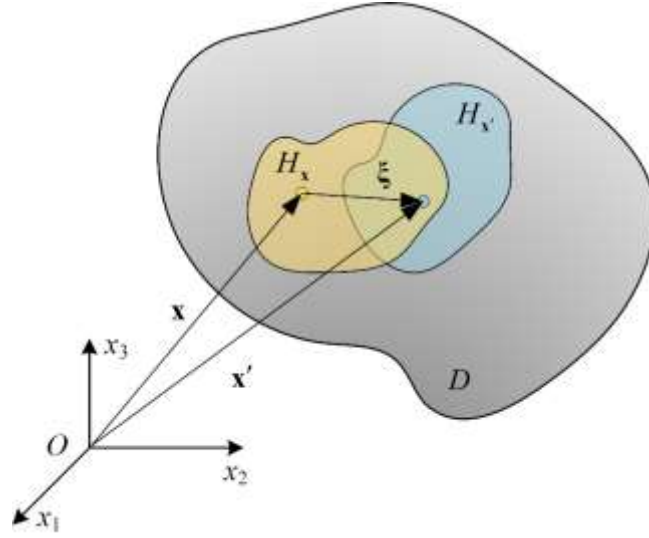


Fig. 1. Interaction of peridynamic points with arbitrary family size and shape.

In order to derive the PD nonlocal expression for the partial derivatives of any order, we assume an orthogonal function, $g_N^{p_1 p_2 \dots p_M}(\xi)$, satisfies the following equation

$$\frac{\partial^{p_1 + p_2 + \dots + p_M} f(\mathbf{x})}{\partial x_1^{p_1} \partial x_2^{p_2} \dots \partial x_M^{p_M}} = \int_{H_x} f(\mathbf{x}') g_N^{p_1 p_2 \dots p_M}(\xi) dV_{x'} \quad (2)$$

where $p_i \in \mathbf{N}, i=1, 2, \dots, M$ and $\sum_{i=1}^M p_i \leq N$. The orthogonal function, also known as PD function, can be constructed as

$$g_N^{p_1 p_2 \dots p_M}(\xi) = \sum_{q_1=0}^N \sum_{q_2=0}^{N-q_1} \dots \sum_{q_M=0}^{N-q_1 \dots q_{M-1}} a_{q_1 q_2 \dots q_M}^{p_1 p_2 \dots p_M} w_{q_1 q_2 \dots q_M}(|\xi|) \xi_1^{q_1} \xi_2^{q_2} \dots \xi_M^{q_M} \quad (3)$$

in which $w_{q_1 q_2 \dots q_M}(|\xi|)$ is the weight function associated with each term $\xi_1^{q_1} \xi_2^{q_2} \dots \xi_M^{q_M}$.

$|\xi|$ is the norm of ξ . $a_{q_1 q_2 \dots q_M}^{p_1 p_2 \dots p_M}$ is the unknown coefficients.

Substituting Eqs. (2) and (3) into Eq. (1) and considering the orthogonality property, one has

$$\sum_{q_1=0}^N \sum_{q_2=0}^{N-q_1} \dots \sum_{q_M=0}^{N-q_1 \dots q_{M-1}} A_{(n_1 n_2 \dots n_M)(q_1 q_2 \dots q_M)} a_{q_1 q_2 \dots q_M}^{p_1 p_2 \dots p_M} = b_{n_1 n_2 \dots n_M}^{p_1 p_2 \dots p_M} \quad (4)$$

where

$$A_{(n_1 n_2 \dots n_M)(q_1 q_2 \dots q_M)} = \int_{H_x} w_{q_1 q_2 \dots q_M}(|\xi|) \xi_1^{n_1+q_1} \xi_2^{n_2+q_2} \dots \xi_M^{n_M+q_M} dV_{x'} \quad (5a)$$

$$b_{n_1 n_2 \dots n_M}^{p_1 p_2 \dots p_M} = n_1! n_2! \dots n_M! \delta_{n_1 p_1} \delta_{n_2 p_2} \dots \delta_{n_M p_M} \quad (5b)$$

in which $n_i \in \mathbf{N}, i=1, 2, \dots, M$, $\sum_{i=1}^M n_i \leq N$. $\delta_{n_i p_i}$ is the Kronecker delta. Eq. (4) can

be also recast in the matrix form as

$$\mathbf{Aa} = \mathbf{b} \quad (6)$$

in which $\mathbf{A}$ is the PD coefficients (shape) matrix, $\mathbf{b}$ is the known coefficients matrix, and $\mathbf{a}$ is the unknown coefficients matrix. After solving Eq. (6) by $\mathbf{a} = \mathbf{A}^{-1} \mathbf{b}$, we can determine the PD function by Eq. (3) and obtained the PD nonlocal expression for the partial derivatives of any order by Eq. (2). It's worth mentioning that the PDDO recovers the local differentiation as the size of family, H_x , decreases or the number of terms in the PD functions, $g_N^{p_1 p_2 \dots p_M}(\xi)$, increases. In addition, the PD functions, $g_N^{p_1 p_2 \dots p_M}(\xi)$, can be constructed for a relative function $f(\mathbf{x}') - f(\mathbf{x})$ and the construction is explained in Appendix A.

3. Thermoelastic steady-state equations for FGM

In the 3-dimensional domain $\Omega \subset \mathbf{R}^3$, the equilibrium equation is stated as follows

$$\sigma_{ij,j} + b_i = 0 \quad (7)$$

where σ_{ij} is the stress tensor, b_i is body force vector.

Hooke's law for an isotropic thermoelastic material within the linear context of small deformation theory is expressed as [2]

$$\sigma_{ij} = \delta_{ij} [\lambda u_{k,k} - \beta(T - T_0)] + \mu(u_{i,j} + u_{j,i}) \quad (8)$$

in which u_i is the displacement vector, T_0 is the reference temperature, δ_{ij} is Kronecker delta. The Lamé constants λ , shear modulus μ and the thermo-stress module β are written as

$$\mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{2\mu\nu}{1-2\nu}, \quad \beta = \frac{E}{1-2\nu}\alpha \quad (9)$$

where E and ν are the Young's modulus and the Poisson's ratio, respectively. α is the coefficient of thermal expansion.

The steady state heat conductive equation is stated as

$$(kT_{,i})_{,i} + \psi = 0 \quad (10)$$

in which T is the temperature, k is the thermal conductivity, ψ is the heat source.

Substituting Eq. (8) into Eq. (7), the equilibrium equation in terms of displacement components can be obtained as follows

$$\left[\lambda u_{k,k} \delta_{ij} + \mu (u_{i,j} + u_{j,i}) \right]_{,j} - \left[\beta (T - T_0) \right]_{,i} + b_i = 0 \quad (11)$$

Eqs. (11) and (10) are the displacement-temperature equations of coupled thermo-elasticity for FGM.

For the sake of simplicity, we take a two-dimensional plane stress problem as an example, but the application of PDDO is not limited to this example. Consider an FGM square plate under mechanical load P and thermal load acting on the left edge, and the coordinate system x_1Ox_2 is established as portrayed in Fig. 2. The length and width of the FGM square plate are W . The effective properties of FGM are continuous functions of the coordinate x_1 as follows

$$E = E(x_1), \quad \nu = \nu(x_1), \quad \alpha = \alpha(x_1), \quad k = k(x_1) \quad (12)$$

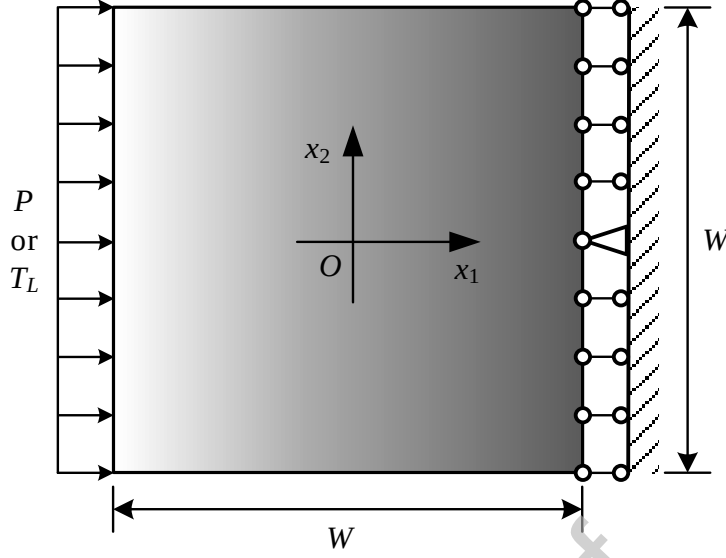


Fig. 2. Schematic diagram of an FGM square plate under mechanical and thermal loads.

Hooke's law for plane stress assumption can be obtained from Eq. (8) as follows

$$\begin{aligned}
 \sigma_{11} &= (\bar{\lambda} + 2\mu) \frac{\partial u_1}{\partial x_1} + \bar{\lambda} \frac{\partial u_2}{\partial x_2} - \bar{\beta}(T - T_0) \\
 \sigma_{22} &= (\bar{\lambda} + 2\mu) \frac{\partial u_2}{\partial x_2} + \bar{\lambda} \frac{\partial u_1}{\partial x_1} - \bar{\beta}(T - T_0) \\
 \sigma_{12} &= \sigma_{21} = \mu \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right)
 \end{aligned} \tag{13}$$

By comparing Eq. (8) with Eq. (13), the Lamé constants λ and the thermo-stress module β are replaced with $\bar{\lambda}$ and $\bar{\beta}$, respectively

$$\bar{\lambda} = \frac{2\mu}{\lambda + 2\mu} \lambda, \quad \bar{\beta} = \frac{2\mu}{\lambda + 2\mu} \beta \tag{14}$$

The displacement–temperature equations of coupled thermo-elasticity for FGM in the absence of the body force and heat source can be rewritten through Eqs. (11), (10), (9) and (12) as

$$\begin{aligned}
& (\bar{\lambda} + 2\mu) \frac{\partial^2 u_1}{\partial x_1^2} + \mu \frac{\partial^2 u_1}{\partial x_2^2} + (\bar{\lambda} + \mu) \frac{\partial^2 u_2}{\partial x_1 \partial x_2} - \bar{\beta} \frac{\partial T}{\partial x_1} + \\
& \left(\frac{d\bar{\lambda}}{dx_1} + 2 \frac{d\mu}{dx_1} \right) \frac{\partial u_1}{\partial x_1} + \frac{d\bar{\lambda}}{dx_1} \frac{\partial u_2}{\partial x_2} - \frac{d\bar{\beta}}{dx_1} (T - T_0) = 0 \\
& (\bar{\lambda} + 2\mu) \frac{\partial^2 u_2}{\partial x_2^2} + \mu \frac{\partial^2 u_2}{\partial x_1^2} + (\bar{\lambda} + \mu) \frac{\partial^2 u_1}{\partial x_1 \partial x_2} - \bar{\beta} \frac{\partial T}{\partial x_2} + \frac{d\mu}{dx_1} \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) = 0 \\
& k \left(\frac{\partial^2 T}{\partial x_1^2} + \frac{\partial^2 T}{\partial x_2^2} \right) + \frac{dk}{dx_1} \frac{\partial T}{\partial x_1} = 0
\end{aligned} \tag{15}$$

We choose the relative function to construct the PD function. Let $M=2$ and $N \geq 2$,

Eq. (A.1) can be written as

$$\frac{\partial^{p_1+p_2} f(x_1, x_2)}{\partial x_1^{p_1} \partial x_2^{p_2}} = \int_{H_x} [f(x'_1, x'_2) - f(x_1, x_2)] g_N^{p_1 p_2}(\xi_1, \xi_2) dA_{x'} \tag{16}$$

where $p_1, p_2 \in \mathbf{N}$ and $1 \leq p_1 + p_2 \leq N$. Substituting Eq. (16) into Eq. (15), the nonlocal form of displacement–temperature equations of coupled thermo-elasticity for FGM can be obtained as follows

$$\begin{aligned}
& \int_{H_x} (u'_1 - u_1) \left[(\bar{\lambda} + 2\mu) g_N^{20} + \mu g_N^{02} + \left(\frac{d\bar{\lambda}}{dx_1} + 2 \frac{d\mu}{dx_1} \right) g_N^{10} \right] dA_{x'} + \\
& \int_{H_x} (u'_2 - u_2) \left[(\bar{\lambda} + \mu) g_N^{11} + \frac{d\bar{\lambda}}{dx_1} g_N^{01} \right] dA_{x'} - \\
& \bar{\beta} \int_{H_x} (T' - T) g_N^{10} dA_{x'} - \frac{d\bar{\beta}}{dx_1} (T - T_0) = 0 \\
& \int_{H_x} (u'_2 - u_2) \left[(\bar{\lambda} + 2\mu) g_N^{02} + \mu g_N^{20} + \frac{d\mu}{dx_1} g_N^{10} \right] dA_{x'} + \\
& \int_{H_x} (u'_1 - u_1) \left[(\bar{\lambda} + \mu) g_N^{11} + \frac{d\mu}{dx_1} g_N^{01} \right] dA_{x'} - \bar{\beta} \int_{H_x} (T' - T) g_N^{01} dA_{x'} = 0 \\
& \int_{H_x} (T' - T) \left[k (g_N^{20} + g_N^{02}) + \frac{dk}{dx_1} g_N^{10} \right] dA_{x'} = 0
\end{aligned} \tag{17}$$

in which

$$u'_1 = u_1(x'_1, x'_2), u'_2 = u_2(x'_1, x'_2), T' = T(x'_1, x'_2), g_N^{p_1 p_2} = g_N^{p_1 p_2}(\xi_1, \xi_2) \tag{18}$$

Similarly, the nonlocal form of Hooke's law, Eq. (13), can be also expressed through

Eq. (16) as

$$\begin{aligned}\sigma_{11} &= (\bar{\lambda} + 2\mu) \int_{H_x} (u'_1 - u_1) g_N^{10} dA_{x'} + \bar{\lambda} \int_{H_x} (u'_2 - u_2) g_N^{01} dA_{x'} - \bar{\beta}(T - T_0) \\ \sigma_{22} &= (\bar{\lambda} + 2\mu) \int_{H_x} (u'_2 - u_2) g_N^{01} dA_{x'} + \bar{\lambda} \int_{H_x} (u'_1 - u_1) g_N^{10} dA_{x'} - \bar{\beta}(T - T_0) \\ \sigma_{12} = \sigma_{21} &= \mu \left\{ \int_{H_x} (u'_1 - u_1) g_N^{01} dA_{x'} + \int_{H_x} (u'_2 - u_2) g_N^{10} dA_{x'} \right\}\end{aligned}\quad (19)$$

As shown in Fig. 2, the left and right edges of the FGM square plate are subjected to constant temperatures of T_L and T_0 , respectively. Also, the plate is subjected to a pressure of P and zero displacement constraints on the left and right edges in x -direction, respectively. Besides, the midpoint of the right edge is subjected to zero displacement constraints in y -direction. The two horizontal edges are insulated and free of tractions. Therefore, the thermal, displacement and stress boundary conditions can be written as

$$\begin{cases} T(-\frac{W}{2}, x_2) = T_L, T(\frac{W}{2}, x_2) = T_0, T_{,2}(x_1, \pm \frac{W}{2}) = 0 \\ \sigma_{22}(x_1, \pm \frac{W}{2}) = \sigma_{12}(x_1, \pm \frac{W}{2}) = \sigma_{12}(\pm \frac{W}{2}, x_2) = 0, -\frac{W}{2} \leq x_1, x_2 \leq \frac{W}{2} \\ \sigma_{11}(-\frac{W}{2}, x_2) = -P, u_1(\frac{W}{2}, x_2) = u_2(\frac{W}{2}, 0) = 0 \end{cases} \quad (20)$$

Substituting Eq. (19) into Eq. (20), the nonlocal forms of boundary conditions are obtained.

4. Numerical implementation

We divide the plate, D , evenly into a finite number of cells with uniform spacing of Δx , each with a specific material point, as shown in Fig. 3. $A = (\Delta x)^2$ stands for the

involved area of material point and $N_{tot}=(W/\Delta x)^2$ represents the total number of material points. The domain $H_{(i)}$ of interaction in the red point $\mathbf{x}_{(i)}$ is defined as a square symmetric family the surrounding blue points form, which can be stated as

$$H_{(i)} := \left\{ \mathbf{x}_{(l)} \in D : |x_{1(l)} - x_{1(i)}| \leq \delta, |x_{2(l)} - x_{2(i)}| \leq \delta \right\} \quad (21)$$

Here shows a specific case of $\delta=3\Delta x$. However, the material points near the boundary, such as the green points $\mathbf{x}_{(j)}$ and $\mathbf{x}_{(k)}$, have nonsymmetric families, $H_{(j)}$ and $H_{(k)}$, respectively. The yellow gradient indicates the influence of the blue points on the central material point, red and green points. In this paper, the weight function associated with each term $\xi_1^{q_1} \xi_2^{q_2}$ is defined as

$$w_{q_1 q_2}(|\xi|) = w(|\xi|) = e^{-2(|\xi|/\delta)^2} \quad (22)$$

where $q_1, q_2 \in \mathbb{N}$ and $1 \leq q_1 + q_2 \leq N$.

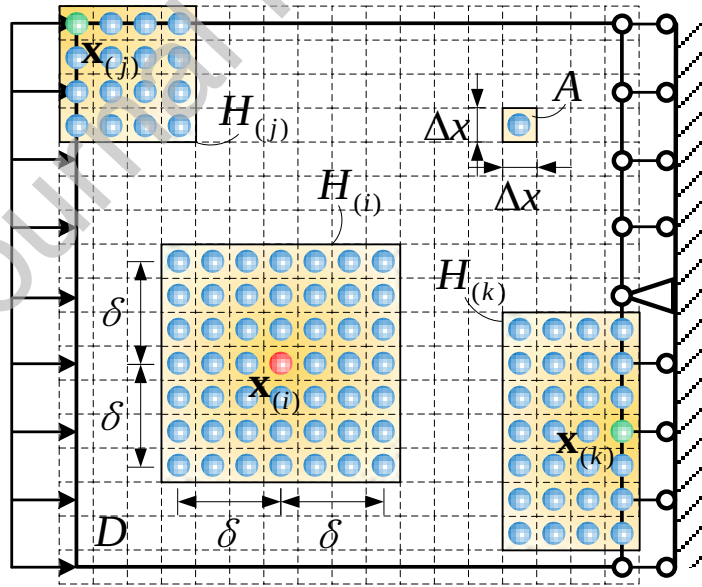


Fig. 3. Uniform PD discretization of a plate with a square family.

As for the boundary conditions, virtual boundary layer needs to be set in PD, but they can act directly on the material points at boundary in PDDO. For example, the

green point $\mathbf{x}_{(j)}$ and $\mathbf{x}_{(k)}$ can be directly subjected to temperatures, displacement and pressure. Hence, the discretized forms of Eqs. (17), (19), (20) and (5a) are written in APPENDIX B.

Since the partial differential equations and boundary conditions are linear, Eqs. (B.1) and (B.4) can be expressed as linear sets of algebraic equations in the form

$$\mathbf{L}\mathbf{u} = \mathbf{b}, \mathbf{c}\mathbf{u} = \mathbf{d} \quad (23)$$

in which, the matrix $\mathbf{L}$ and $\mathbf{c}$ denote the coefficients arising from the PD differentiation for Eqs. (B.1) and (B.4), respectively. The vector $\mathbf{b}$ contains the known values at each point and here $\mathbf{b}=\mathbf{0}$. The vector $\mathbf{d}$ includes the specified known values of Eq. (B.4) at boundary points. The vector $\mathbf{u}$ contains the PD unknowns and can be written as

$$\mathbf{u} = \begin{bmatrix} u_{1(1)} & u_{2(1)} & T_{(1)} & u_{1(2)} & u_{2(2)} & T_{(2)} & \cdots & u_{1(N_{tol})} & u_{2(N_{tol})} & T_{(N_{tol})} \end{bmatrix}^T \quad (24)$$

By introducing the Lagrange multiplier in vector, λ , and employing the concept of variational analysis, Eq. (23) can be combined as

$$\delta\mathbf{u}^T (\mathbf{L}\mathbf{u} - \mathbf{b}) + \delta[\lambda^T (\mathbf{c}\mathbf{u} - \mathbf{d})] = 0 \quad (25)$$

Expanding the first variation of the second term in Eq. (25) yields

$$\delta\mathbf{u}^T (\mathbf{L}\mathbf{u} - \mathbf{b}) + \delta\lambda^T (\mathbf{c}\mathbf{u} - \mathbf{d}) + \delta\mathbf{u}^T \mathbf{c}^T \lambda = 0 \quad (26)$$

For arbitrary variations of $\delta\mathbf{u}$ and $\delta\lambda$, Eq. (26) leads to

$$\mathbf{L}\mathbf{u} - \mathbf{b} + \mathbf{c}^T \lambda = \mathbf{0}, \mathbf{c}\mathbf{u} - \mathbf{d} = \mathbf{0} \quad (27)$$

Eq. (27) can be recast as the system of algebraic equations as follows

$$\begin{bmatrix} \mathbf{L} & \mathbf{c}^T \\ \mathbf{c} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \mathbf{u} \\ \lambda \end{Bmatrix} = \begin{Bmatrix} \mathbf{b} \\ \mathbf{d} \end{Bmatrix} \quad (28)$$

We use the Intel direct solver PARDISO to achieve the solution to Eq. (28). Then, in accordance with the solution, $\mathbf{u}$, and Eq. (B.3), the stress components can be also obtained by calculation.

5. Numerical examples

In this section, four numerical simulations are implemented by using the PDDO modelling. The first problem is a homogeneous plate with hole in tension and the convergence of the error in a specific norm for under difference discretizations is tested by comparison with the analytical solution. The second problem is bending of two kinds of FGM beams under mechanical load and the PDDO solutions are compared with the results from Ref. [51-53]. The third problem is a homogeneous plate under mechanical and thermal load and a comparison is carried out with the FEM results. The fourth problem is an FGM plate under mechanical and thermal loads and the effective properties are estimated by the Mori-Tanaka method. The fifth problem is an FGM plate with a crack under thermal load. In the following simulation, the horizon size and the number of terms in the TSE are specified as $\delta=3\Delta x$ and $N=2$, respectively, unless otherwise stated. In the third, fourth, fifth problems, the square plate length and the uniform spacing are respectively fixed at $W=1\text{m}$ and $\Delta x=5\text{mm}$ and for convenience, the dimensionless coordinate, $\bar{x}_1$, $\bar{x}_2$, displacements, $\bar{u}_1$, $\bar{u}_2$, stresses, $\bar{\sigma}_{11}$, $\bar{\sigma}_{22}$, $\bar{\sigma}_{12}$, and temperature $\bar{T}$ are defined by

$$\bar{x}_1 = \frac{x_1}{W}, \bar{x}_2 = \frac{x_2}{W}, \bar{u}_1 = \frac{u_1}{u^*}, \bar{u}_2 = \frac{u_2}{u^*}, \bar{\sigma}_{11} = \frac{\sigma_{11}}{\sigma^*}, \bar{\sigma}_{22} = \frac{\sigma_{22}}{\sigma^*}, \bar{\sigma}_{12} = \frac{\sigma_{12}}{\sigma^*}, \bar{T} = \frac{T}{T^*} \quad (29)$$

where the scale factors are $u^*=0.1\text{mm}$, $\sigma^*=10\text{MPa}$, $T^*=100\text{K}$.

5.1. A homogeneous plate with hole in tension

Consider a homogeneous infinite plate with hole in tension, as shown in Fig. 4.

Because of symmetry, one quarter of the plate is analyzed by fixing the boundary $x=0$ ($y=0$) in x -direction (y -direction) and the discretization for the uniform spacing $\Delta x=0.1$ is shown in Fig. 5. The length of the plate is 5 and the radius of the hole is $a = 1$. The Young's modulus and the Poisson's ratio are $E=1000$ and $\nu=0.3$. For plane stress conditions, the exact displacement and stress in Cartesian coordinates [41] can be expressed as

$$\begin{aligned} u_1(r, \theta) &= \frac{Pa}{8\mu} \left\{ \frac{r}{a} (\kappa+1) \cos \theta + \frac{2a}{r} [(1+\kappa) \cos \theta + \cos 3\theta] - \frac{2a^3}{r^3} \cos 3\theta \right\}, \\ u_2(r, \theta) &= \frac{Pa}{8\mu} \left\{ \frac{r}{a} (\kappa-3) \sin \theta + \frac{2a}{r} [(1-\kappa) \sin \theta + \sin 3\theta] - \frac{2a^3}{r^3} \sin 3\theta \right\}, \\ \sigma_{11}(r, \theta) &= P - P \frac{a^2}{r^2} \left(\frac{3}{2} \cos 2\theta + \cos 4\theta \right) + P \frac{3a^4}{2r^4} \cos 4\theta, \\ \sigma_{22}(r, \theta) &= -P \frac{a^2}{r^2} \left(\frac{1}{2} \cos 2\theta - \cos 4\theta \right) - P \frac{3a^4}{2r^4} \cos 4\theta, \\ \sigma_{12}(r, \theta) &= -P \frac{a^2}{r^2} \left(\frac{1}{2} \sin 2\theta - \sin 4\theta \right) - P \frac{3a^4}{2r^4} \sin 4\theta \end{aligned} \quad (30)$$

Where $\kappa = (3-\nu)/(1+\nu)$.

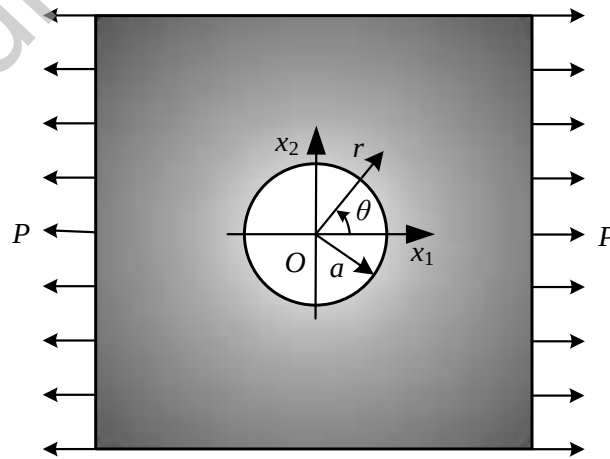


Fig. 4. Setup of the homogeneous plate with a hole in tension.

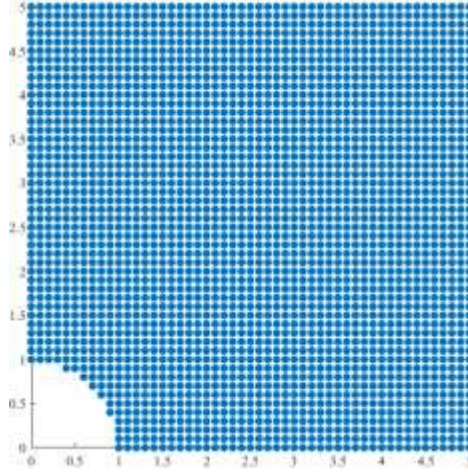


Fig. 5. Discretization of one quarter of the plate for $\Delta x=0.1$

The boundary points on $x=5$ and $y=5$ are subjected to the traction force calculated by Eq. (30). Five discretization cases with the uniform spacing, $\Delta x=0.1, 0.05, 0.04, 0.025, 0.02$, are tested, respectively. The displacement and stress on polar coordinate $r=2a$ for the uniform spacing $\Delta x=0.04$ are compared with the exact solutions, as shown in Fig. 6. The L2-norm [41] is calculated by

$$\|\mathbf{u}\|_{L2} = \sqrt{\frac{\sum_{i=1}^{N_{tot}} (\mathbf{u}_{(i)} - \mathbf{u}_{(i)}^{exact}) \cdot (\mathbf{u}_{(i)} - \mathbf{u}_{(i)}^{exact}) A}{\sum_{i=1}^{N_{tot}} \mathbf{u}_{(i)}^{exact} \cdot \mathbf{u}_{(i)}^{exact} A}} \quad (31)$$

The convergence of the L2-norm for the displacement field of plate with hole in tension is plotted in Fig. 7. The comparison in Fig. 6 demonstrates the accuracy of the present PDDO modeling in solving the deformation of elastic solid and it can be seen from Fig. 7 the present PDDO modeling has good convergence.

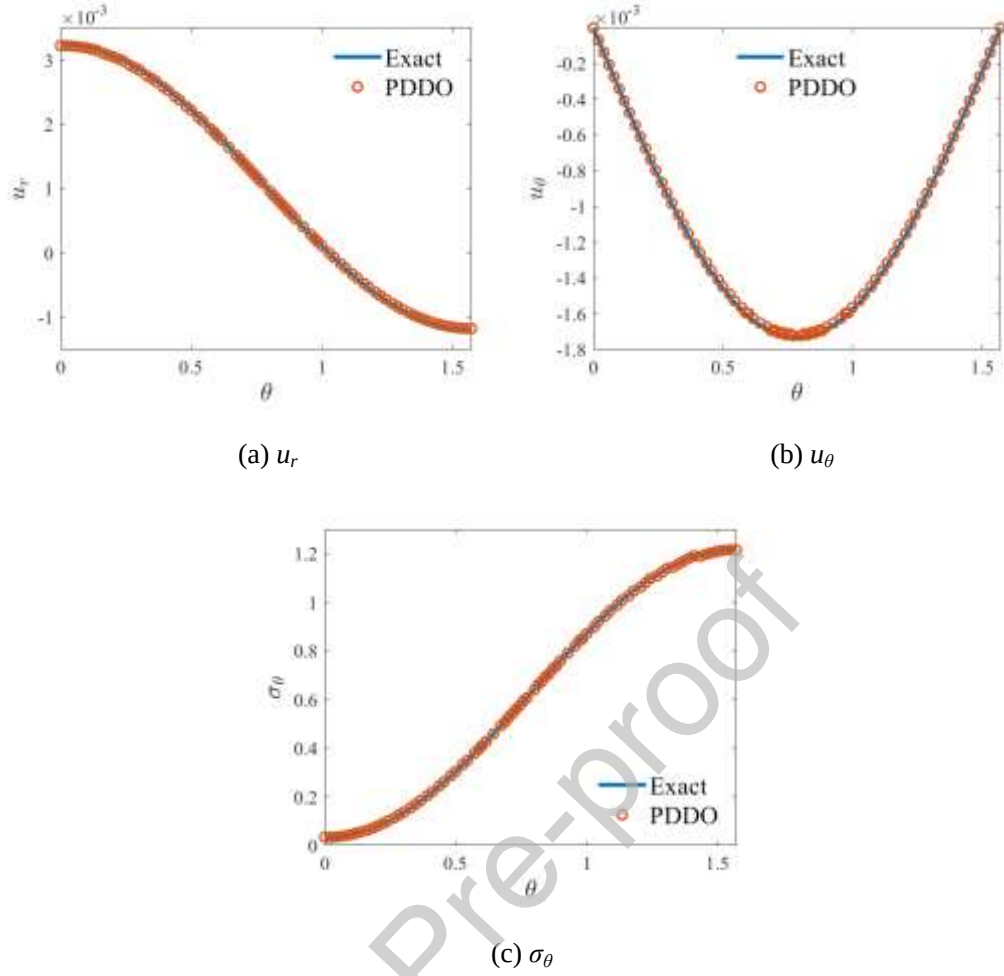


Fig. 6. Displacement and stress for points on $r=2a$ for $\Delta x=0.04$.

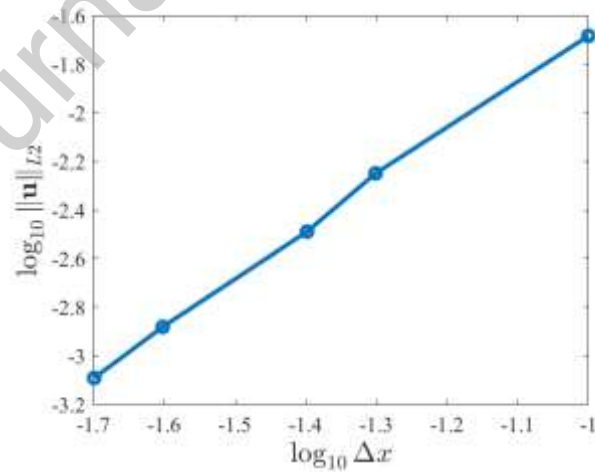


Fig. 7. Convergence of the L_2 -norm for the displacement of plate with hole in tension.

5.2. Bending of FGM beams under mechanical loads

Compare with Ying et al.'s study [51] of simply supported FGM beams resting on a two-parameter elastic foundation, and the coordinate system x_1Ox_2 is built as shown in Fig. 8. The length and thickness of beam are L and H and the depth-to-length ratio is fixed at $H/L=1/15$. The Poisson's ratio is a constant of $\nu=0.3$ and the Young's modulus is an exponent-law distribution function of thickness as follows

$$E = E_0 e^{\gamma x_2} \quad (32)$$

where E_0 is the value of the Young's modulus at $x_2 = 0$, and γ is the gradient index.

The boundary conditions can be given as follows

$$\begin{aligned} \sigma_{11} = u_2 = 0, \text{ at } x_1 = 0, L, 0 \leq x_2 \leq H \\ \sigma_{12} = 0, \sigma_{22} = q(x_1), \text{ at } 0 \leq x_1 \leq L, x_2 = H \\ \sigma_{12} = \sigma_{22} - (K_w u_2 - K_p \frac{\partial^2 u_2}{\partial x_1^2}) = 0, \text{ at } 0 \leq x_1 \leq L, x_2 = 0 \end{aligned} \quad (33)$$

in which $q(x_1)$ is the transverse load on the upper surface and the sinusoidal distributed load is taken as $q(x_1) = -q_0 \sin(\pi x_1/L)$ with $q_0 = 10^6$. K_w and K_p are the Winkler elastic coefficient and shearing layer coefficient of foundation and their dimensionless form $\bar{K}_w$ and $\bar{K}_p$ are defined as

$$\bar{K}_w = \frac{12K_w L^4}{E_0 H^3}, \bar{K}_p = \frac{12K_p L^2}{E_0 H^3} \quad (34)$$

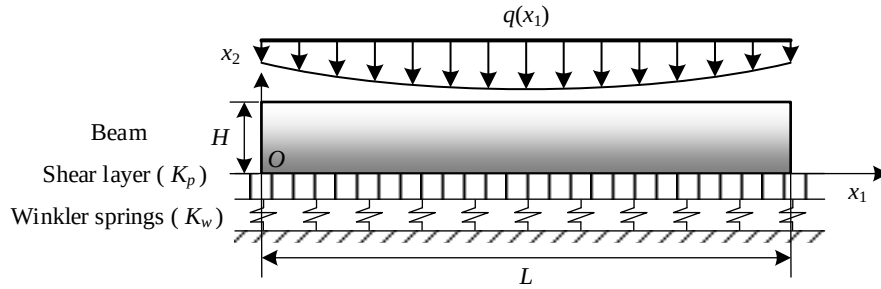


Fig. 8. An FGM beam resting on a two-parameter elastic foundation.

The uniform spacing is specified as $\Delta x = 0.02H$. The material parameters in two different cases are analysed. In Case 1, $E_0 = 100\text{GPa}$, $\gamma = \ln 10$, and in Case 2, $E_0 = 1000\text{GPa}$, $\gamma = -\ln 10$. Fig. 9 provides a comparison of results for different foundation parameters of $\bar{K}_w = 0, 100$ and $\bar{K}_p = 0, 25$. As shown in Fig. 9, the curves without circles are the results in Case 1, and those with circles in Case 2. The black curves are the solutions obtained in Ref. [51], and the red curves are the PDDO solutions. It is evident that the red and black curves coincide almost exactly and the PDDO solutions agree well with the solutions given by Ying et al[51]. It verifies the capability of the present PDDO modelling in simulating FGM structure on the elastic foundation.

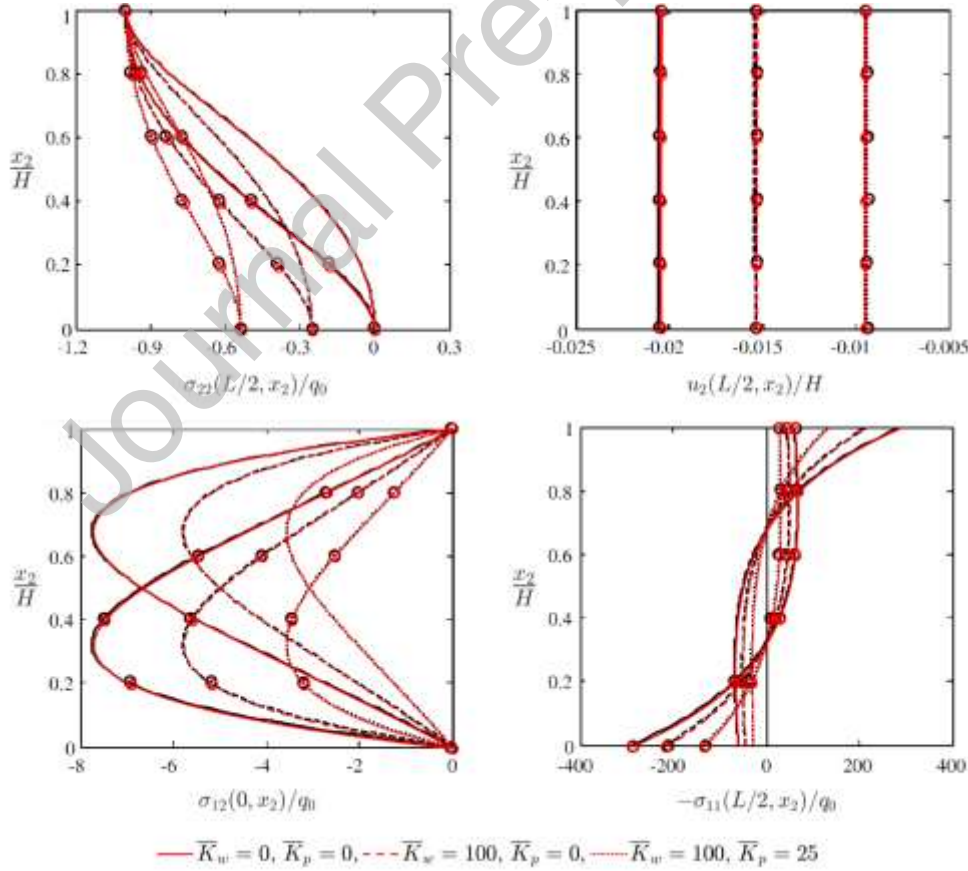


Fig. 9. Transverse displacement and stresses distributions along the thickness direction: the curves without circle for Case 1, and those with circles for Case 2; black curves are solutions from Ref. [51], and red curves are PDDO solutions.

An FGM beam with parabolic concave lower surface is depicted in Fig. 10. The total length is $L = 200\text{mm}$ and the depth-to-length ratios at the two ends are $H/L=0.1$. The mid-span thickness is H_1 , and the lower surface of beam can be described as $h(x)=(H_1-H)(x-100)^2/10^4 - H_1$. The Poisson's ratio is also fixed at $\nu=0.3$. The Young's modulus distribution is the same as Eq. (29) with $E_0 = 2.2 \times 10^5 \text{MPa}$ and $\gamma = -0.05$. The boundary conditions can be written as follows

$$\begin{aligned} \sigma_{11} = u_2 = 0, & \text{ at } x_1 = 0, L, -H \leq x_2 \leq 0 \\ \sigma_{12} = 0, \sigma_{22} = q(x_1), & \text{ at } 0 \leq x_1 \leq L, x_2 = 0 \\ l\sigma_{11} + m\sigma_{12} = m\sigma_{22} + l\sigma_{12} = 0, & \text{ at } 0 \leq x_1 \leq L, x_2 = h(x_1) \end{aligned} \quad (35)$$

where the uniform transverse load is taken as $q(x_1)=-100\text{N/mm}$. l and m are the direction cosine the outer normal of the lower surface

$$l = l(x_1) = \frac{dh(x_1)}{dx_1} / \sqrt{1 + \left[\frac{dh(x_1)}{dx_1}\right]^2}, \quad m = m(x_1) = -1 / \sqrt{1 + \left[\frac{dh(x_1)}{dx_1}\right]^2} \quad (36)$$

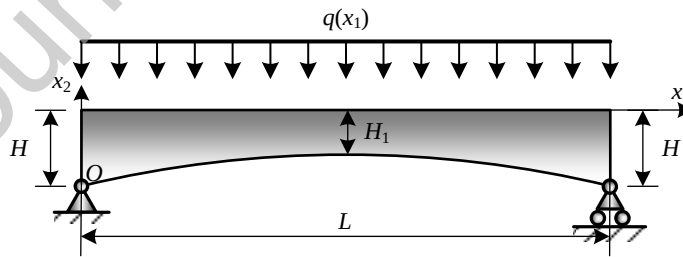


Fig. 10. An FGM beam with parabolic concave lower surface.

The uniform spacing is specified as $\Delta x=0.02H$. The deformed configurations for the beams are calculated for three different depth ratios $H_1/H=0.5, 0.7, 0.9$, as shown in Fig.11. The smoothed particle hydrodynamics (SPH) results and FEM results given in Ref. [52] are plotted in Fig. 11(a) and (b). The analytical results obtained from Ref.

[53] are portrayed in Fig. 11(c) and the PDDO results are shown in Fig. 11(d). It can be seen from Fig. 11 that the relative errors of the maximal transverse displacement for three different depth ratios between the SPH and analytical results are 0.35%, 2.61% and 4.23%. The relative errors between the FEM and analytical solutions are 4.46%, 5.86% and 5.95%. By comparison, the relative errors of PDDO results are only 0.03%, 0.049% and 0.053%. It indicates that the PDDO solutions match better with the analytical results than SPH and FEM results. This comparison study validates the accuracy of the present PDDO modelling in solving FGM structure with curved boundaries.

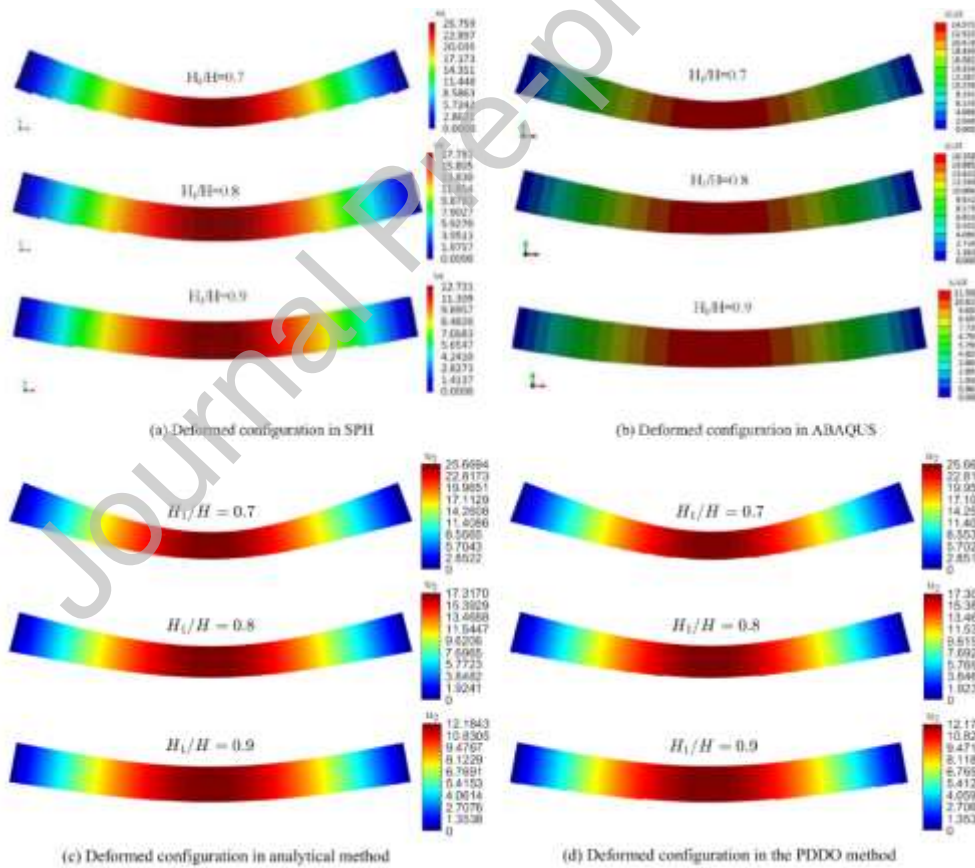


Fig. 11. Deformed configurations of the FGM beams with parabolic concave lower surface ((a), (b) from Ref. [52], (c) from Ref. [53], (d) shows PDDO solutions).

5.3. A homogeneous plate under mechanical and thermal load

Consider a homogeneous plate under mechanical thermal load, as portrayed in Fig. 2. The Young's modulus is $E=200\text{GPa}$. The Poisson's ratio is also fixed at $\nu=0.3$. The coefficient of thermal expansion is $\alpha=16\times 10^{-6}/\text{K}$. The thermal conductivity is $k=20\text{W}/(\text{m}\cdot\text{K})$. The reference temperature is $T_0=20\text{K}$. The boundary conditions are the same as Eq. (20). The left edge of the square plate is subjected to a constant temperature of $T_L=100\text{K}$ and a pressure of $P=10\text{MPa}$.

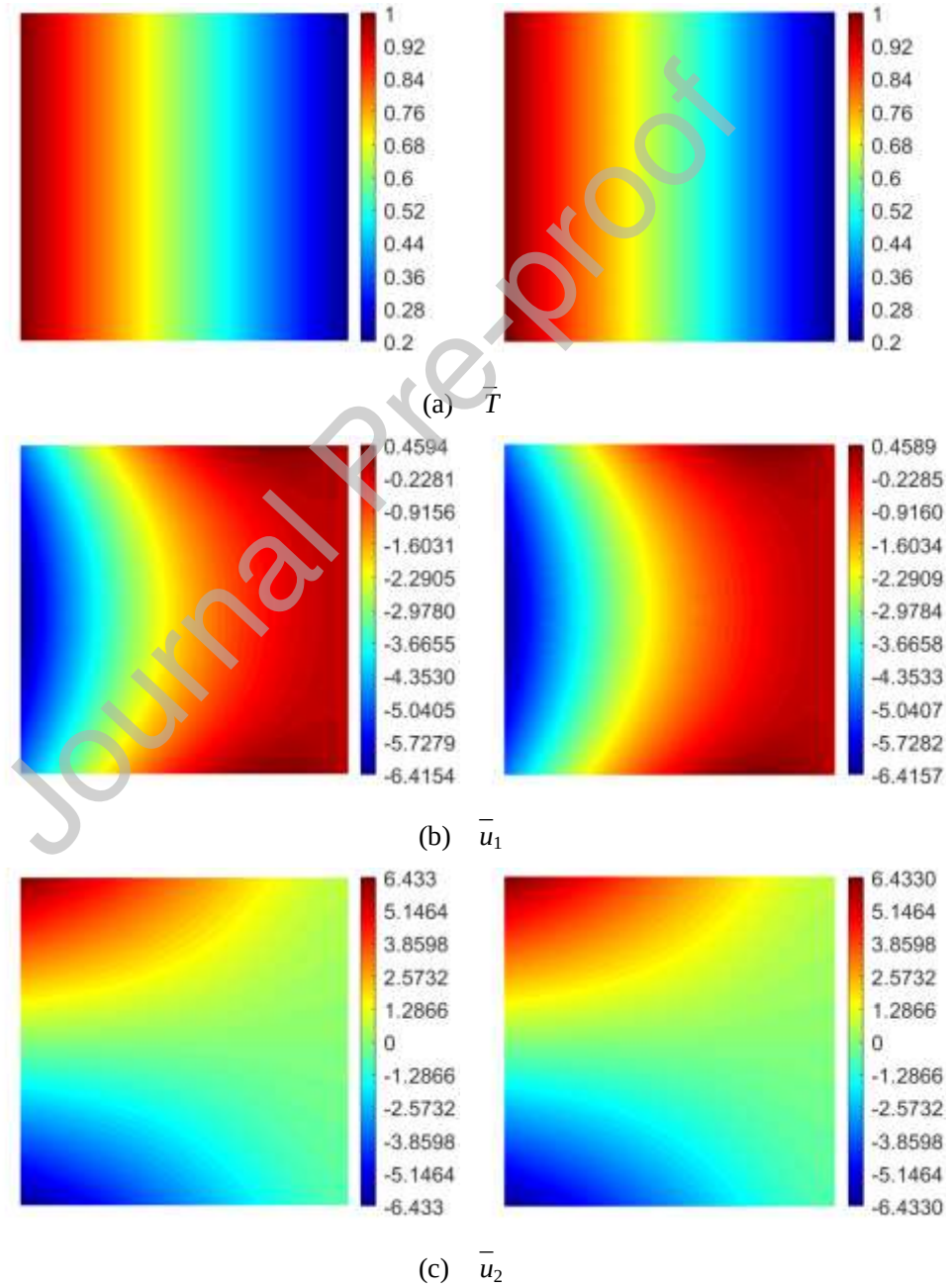
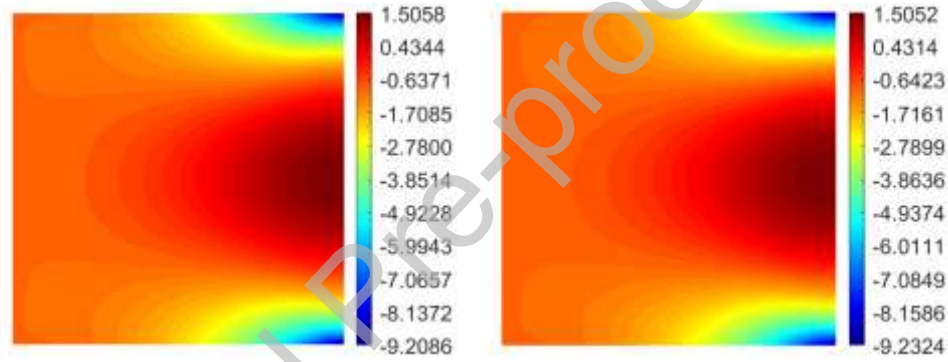
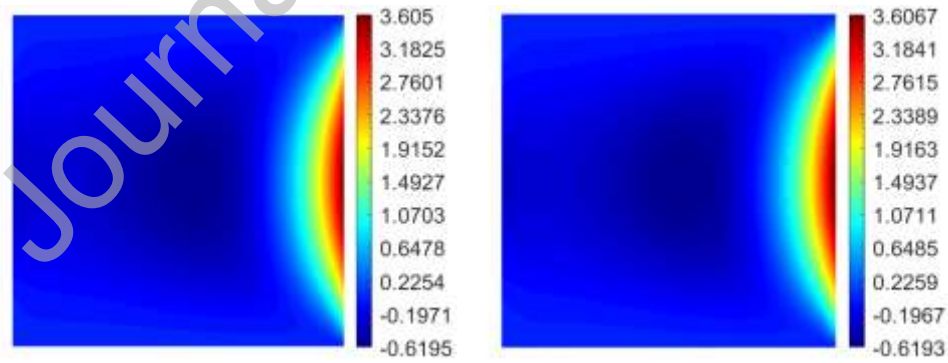


Fig. 12. Temperature and displacement contours (Left: FEM results, right: PDDO solutions).

Figs. 12 and 13 present the comparison of the temperature, displacement and stress contours predicted by the PDDO modeling against the FEM, and the FEM and PDDO results are placed in left and right, respectively. The plate is meshed with 50×50 8-node quadrilateral 2D elements in finite element simulation. As evident in Figs. 12 and 13, the PDDO and FEM solutions are in excellent agreement. This comparison study demonstrates the accuracy of the present PDDO modeling in solving the thermoelasticity problem.



(a) $\bar{\sigma}_{11}$



(b) $\bar{\sigma}_{22}$

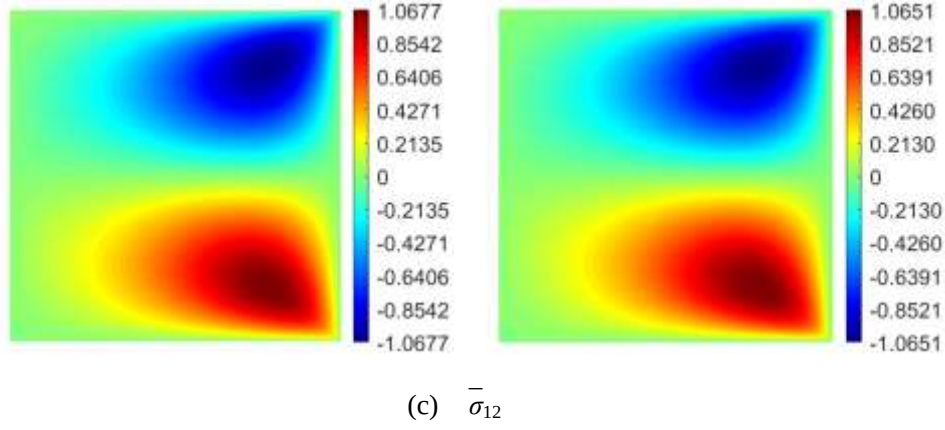


Fig. 13. Stress contours of the homogeneous plate (Left: FEM results, right: PDDO solutions).

5.4. An FGM plate under mechanical and thermal loads

We consider an FGM plate under mechanical and thermal loads, as shown in Fig. 2. The FGM of this plate is fabricated by mixing ceramic, zirconia (ZrO_2), and metal, Monel (70Ni-30Cu), which is widely applied in aircraft engines. For this two-phase composite material, we choose the Mori-Tanaka method to estimate the effective properties, and metal and ceramic serve as a particulate phase and a matrix phase, respectively. According to Ref. [54,55], the effective bulk modulus K , shear modulus μ , heat conduction coefficient k , coefficient of thermal expansion α are given as

$$\begin{aligned}
 \frac{K - K_m}{K_c - K_m} &= V_c \left/ \left[1 + (1 - V_c) \frac{K_c - K_m}{K_m + 4\mu_m/3} \right] \right., \\
 \frac{\mu - \mu_m}{\mu_c - \mu_m} &= V_c \left/ \left[1 + (1 - V_c) \frac{\mu_c - \mu_m}{\mu_m + f_m} \right] \right. \text{ with } f_m = \frac{\mu_m(9K_m + 8\mu_m)}{6(K_m + 2\mu_m)}, \\
 \frac{k - k_m}{k_c - k_m} &= V_c \left/ \left[1 + (1 - V_c) \frac{k_c - k_m}{3k_m} \right] \right., \\
 \frac{\alpha - \alpha_m}{\alpha_c - \alpha_m} &= \left(\frac{1}{K} - \frac{1}{K_m} \right) \left/ \left(\frac{1}{K_c} - \frac{1}{K_m} \right) \right.
 \end{aligned} \tag{37}$$

where the indices c and m represent ceramic and metallic phases, respectively. Note that the Lamé constants λ and the thermo-stress module β also have the correspondence relations of $\lambda = K - 2\mu/3$ and $\beta = 3K\alpha$, respectively. V_c denote the volume

fraction of the particulate phase, or ceramic phase, that is assumed to be the power-law-type function as follows

$$V_c = (0.5 - \bar{x}_1)^{n_g} \quad (38)$$

in which n_g is a parameter that dictates the different material gradients. The volume fraction of the metallic phase is $V_m = 1 - V_c$. The properties of these two material phases, zirconia and metal, are given by [54,55]

$$\begin{aligned} K_c &= 125.83 \text{ GPa}, \mu_c = 58.08 \text{ GPa}, \alpha_c = 10 \times 10^{-6} / \text{K}, k_c = 2.09 \text{ W / mK}, \text{ for zirconia} \\ K_m &= 227.24 \text{ GPa}, \mu_m = 65.55 \text{ GPa}, \alpha_m = 15 \times 10^{-6} / \text{K}, k_m = 25 \text{ W / mK}, \text{ for Monel} \end{aligned} \quad (39)$$

The reference temperature is $T_0 = 300 \text{ K}$. The boundary conditions are the same as Eq. (20) and two type loading conditions are imposed as

Thermal loading

$$P = 0, T_L = 1000 \text{ K} \quad (40)$$

Combined thermal and pressure loading

$$P = 100 \text{ MPa}, T_L = 1000 \text{ K} \quad (41)$$

Fig. 14 shows the horizontal distributions of displacements, stresses and temperature for three different material gradients $n_g = 1, 5, 10$ under the thermal loading. Since the symmetry of geometric configuration, material properties and boundary conditions, the variations of vertical displacement $\bar{u}_2$ and shear stress $\bar{\sigma}_{12}$ are chosen at $\bar{x}_2 = 0.25$ in Fig. 14, but others are selected at $\bar{x}_2 = 0$. It can be seen from Fig. 14 that increase occurs for the horizontal and vertical displacements, $\bar{u}_1$, $\bar{u}_2$, and maximal horizontal normal stress, $\bar{\sigma}_{11}$, as the material gradient, n_g , increases. It's worth noting that the distributions of vertical normal stress, $\bar{\sigma}_{22}$, and shear stress, $\bar{\sigma}_{12}$, are greatly

influenced by the material gradient, n_g . The distributions of normal stress, $\bar{\sigma}_{11}$, and shear stress, $\bar{\sigma}_{12}$, in $-0.5 < \bar{x}_1 < -0.25$ even appear in the opposite direction with the increase of the material gradient, n_g .

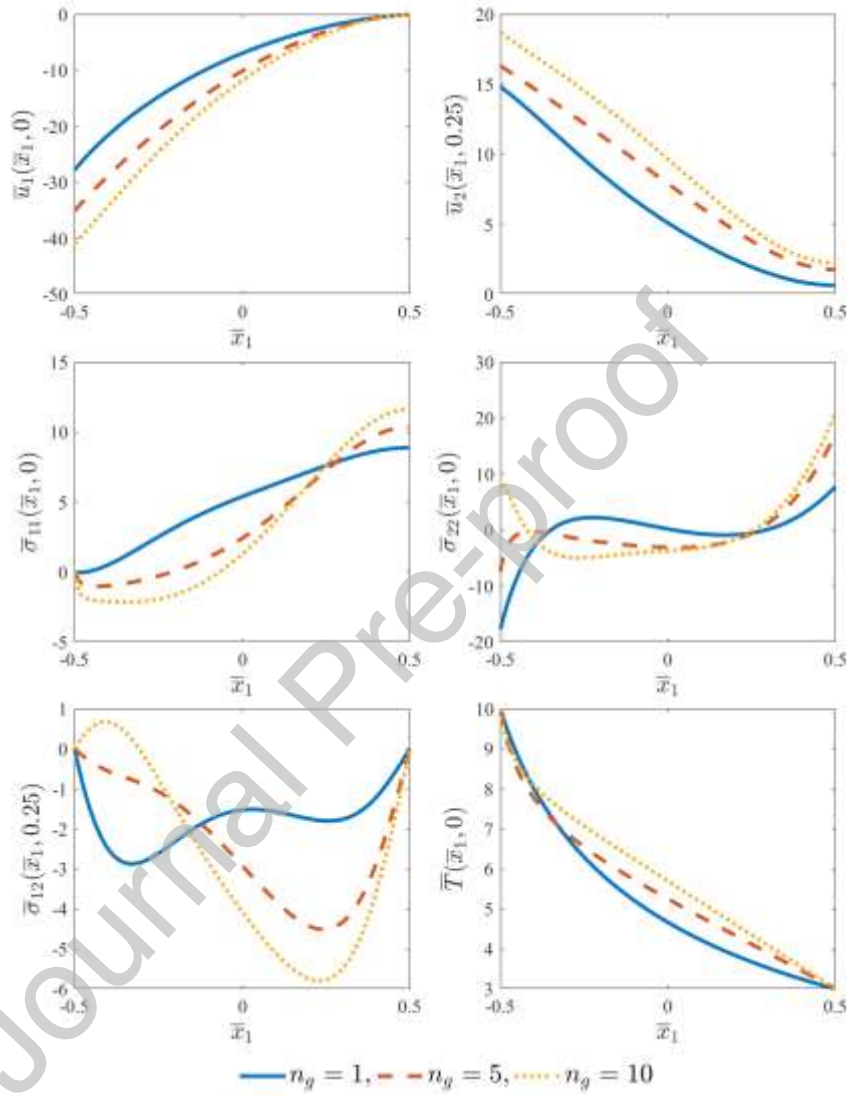


Fig. 14. Horizontal distributions of displacements, stresses and temperature for different material gradients under the thermal loading.

The temperature $\bar{T}$ presents a nonlinear distribution due to the change of material gradient, and the closer the volume fractions of two material phases are, the more obvious the nonlinear distribution will be. Fig. 15 shows the horizontal distributions of displacements, stresses and temperature for three different material gradients

$n_g=1,5,10$ under the combined thermal and pressure loading. Comparing the solutions in Figs. 14 and 15, it can be found that the trends of displacements, stresses and temperature for three different material gradients under different boundary conditions are consistent. By observing the values in two figures, it is derived that as a consequence of the application of pressure loading, the horizontal displacement, $\bar{u}_1$, is scaled down by a certain amount, and the horizontal normal stress, $\bar{\sigma}_{11}$, goes down by P , but others have little change.

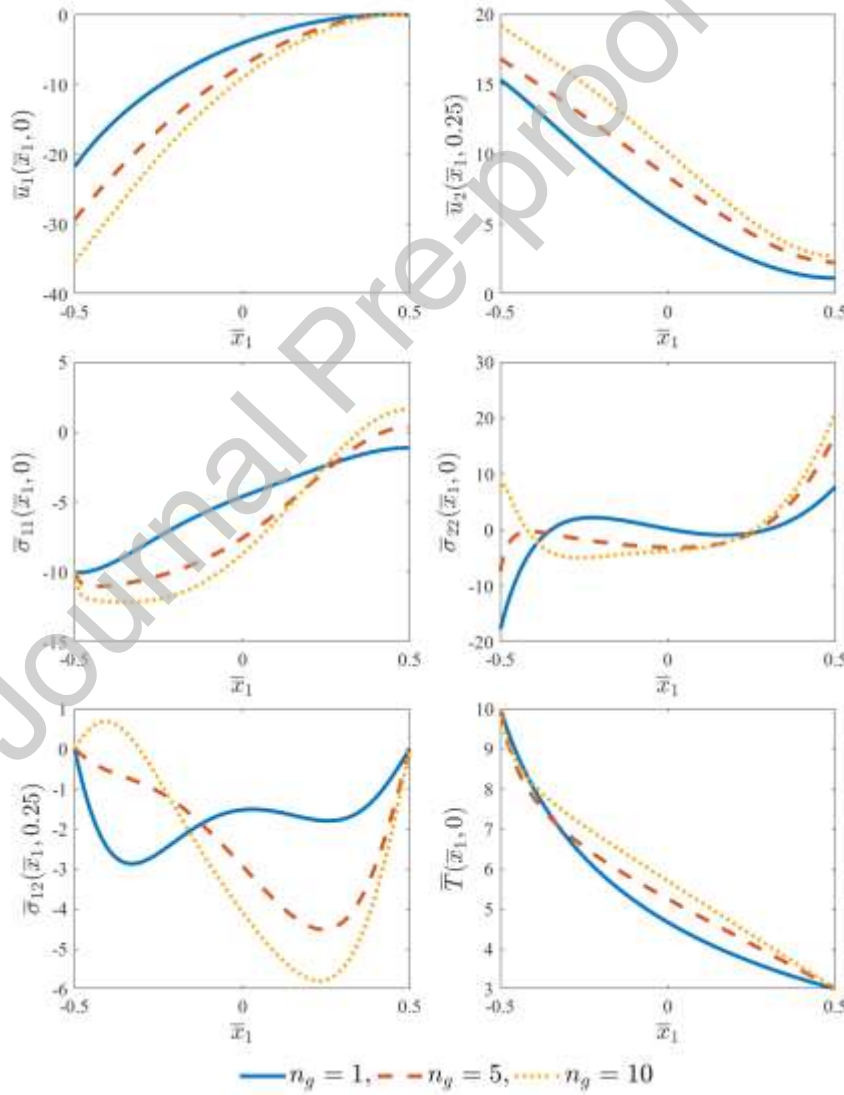


Fig. 15. Horizontal distributions of displacements, stresses and temperature for different material gradients under the combined thermal and pressure loading.

5.5. An FGM plate with a pre-existing crack under thermal load

We add a crack to the FGM plate discussed in the section 5.3, as shown in Fig. 16, and consider the thermal loading, Eq. (37), the material gradients, $n_g=1$. The crack (red thread in Fig. 16) locates at $x_1=-\Delta x/2$ and the crack length is fixed at $2a=35\Delta x$. As illustrated in Fig. 16, the point will be removed from the family members, $H_{(i)}$, of the red point, $x_{(i)}$, if they are located across from the crack, like the gray point. We assume that the both sides of the crack are insulated and free of tractions. Therefore, the boundary conditions on both sides of the crack can be expressed as

$$\sigma_{11(k)} = \sigma_{12(k)} = 0, T_{,1(k)} = 0, \text{ for } x_{1(k)} = 0, -\Delta x, \text{ and } -17.5\Delta x \leq x_{2(k)} \leq 17.5\Delta x \quad (42)$$

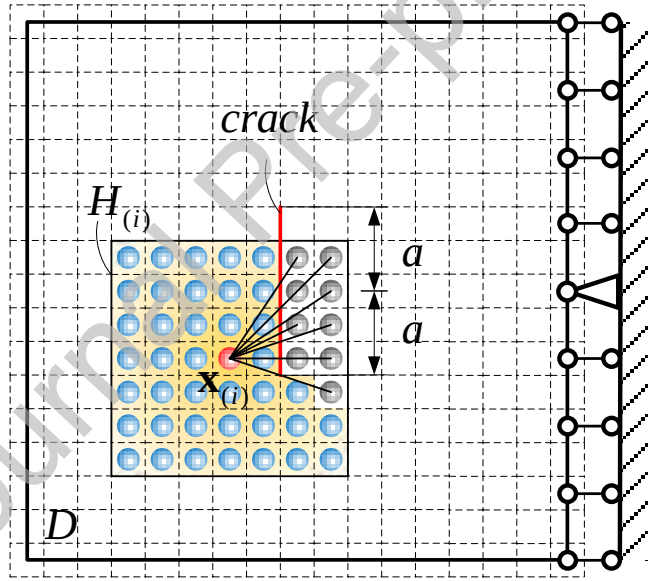


Fig. 16. Description of a crack by removing the interactions across the crack.

Fig. 17 shows the distributions of stress and temperature field in deformed configurations of an FGM plate under the thermal loading and the deformation is magnified by 50 times. It is obvious from Fig. 17 that the existence of cracks leads to discontinuity of displacements, $\bar{u}_1$, $\bar{u}_2$ and temperature, $\bar{T}$, and singularity of stresses, $\bar{\sigma}_{11}$, $\bar{\sigma}_{22}$, $\bar{\sigma}_{12}$. To further study the singularity of stress, we investigate the

vertical distributions of horizontal normal stress, $\bar{\sigma}_{11}$, and shear stress, $\bar{\sigma}_{12}$, near the crack tip for three different horizon sizes, $\delta=2\Delta x$, $3\Delta x$, $5\Delta x$, and two different numbers of terms in the TSE, $N=2, 3$, as shown in Figs. 18 and 19, respectively. Note that the black horizontal line marks the location of crack tip. It can be seen from Figs. 18 and 19 that with the decreasing of horizon size, δ , or the increasing of number of terms in the TSE, N , the horizontal normal stress, $\bar{\sigma}_{11}$, and shear stress, $\bar{\sigma}_{12}$, near the crack tip approach a singular behavior. The degree of nonlocality reduces with decreasing family size and with increasing number of terms in the TSE.

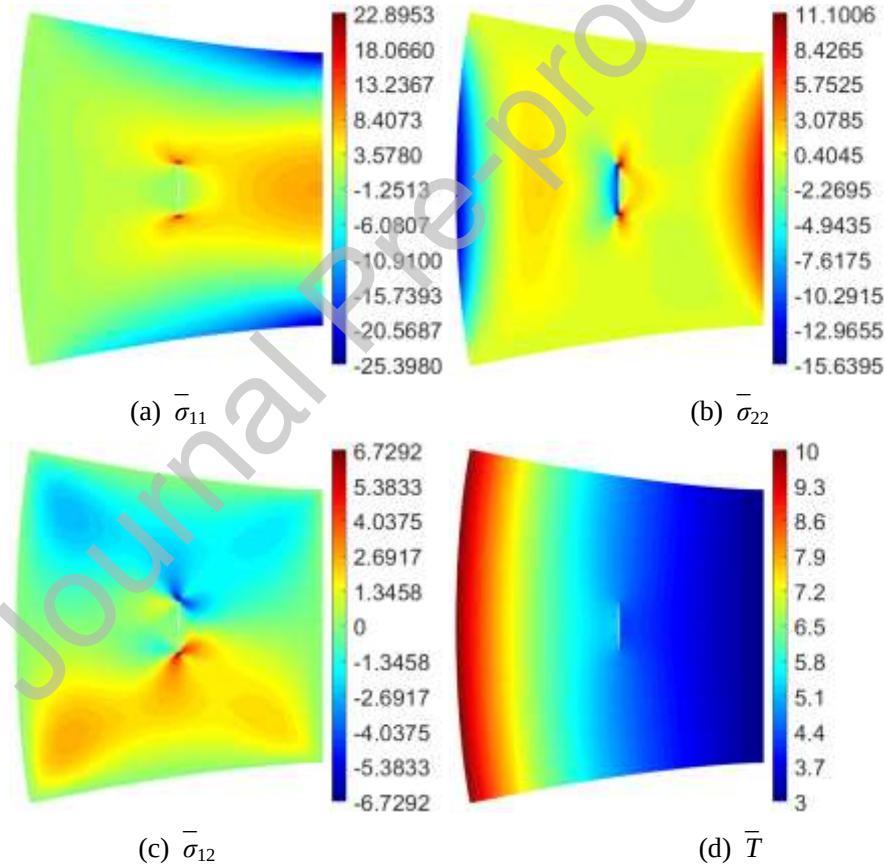


Fig. 17. Distributions of stress and temperature field in deformed configurations of an FGM plate under thermal loading.

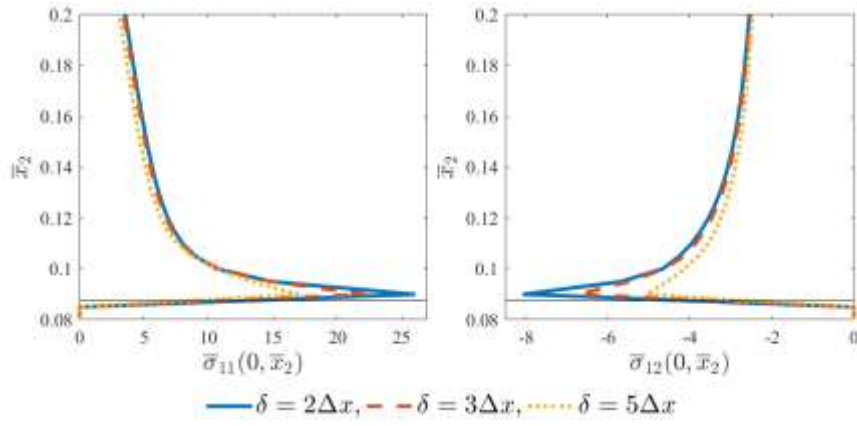


Fig. 18. Vertical distributions of stresses near the crack tip for different horizon size (The black horizontal line marks the location of crack tip).

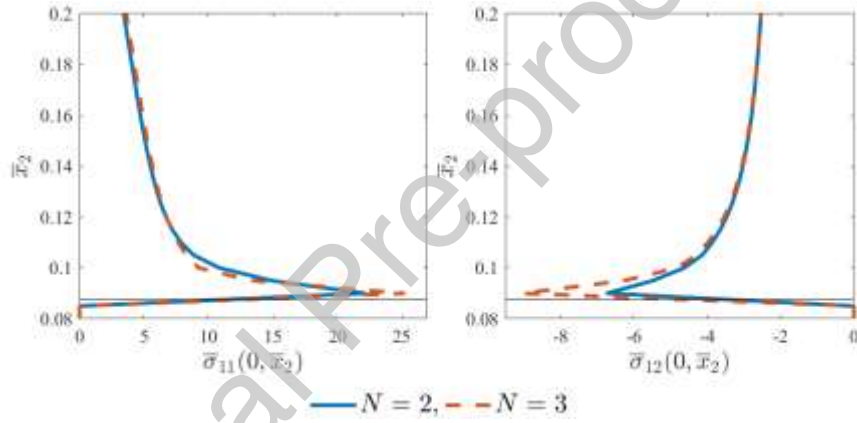


Fig. 19. Vertical distributions of stresses near the crack tip for different number of terms in the TSE (The black horizontal line marks the location of crack tip).

6. Conclusions

A nonlocal model for steady-state thermoelastic analysis of FGMs is developed by using peridynamic differential operator in this paper. A two-dimensional FGM square plate under mechanical and thermal loads is taken as an example. The Hooke's law for isotropic thermoelastic materials, displacement-temperature equations of coupled thermoelasticity, and boundary conditions are reformulated from their local

differential forms into their nonlocal integral forms. The temperature, displacement and stress fields are solved by introducing the Lagrange multiplier, employing variational analysis, and using the Intel direct solver PARDISO. The accuracy and convergence of this model is demonstrated by the good agreements between the predicted solutions and the FEM solutions and the analytical solutions in the previously published literature. Finally, by using the Mori-Tanaka method to estimate the effective properties, and the effects of different material gradients and loads on the physical fields of the FGM plate are investigated. Furthermore, an initial crack is located in the FGM plate, and the effect of the degree of nonlocality on the stress singularity of the crack tip is examined further. The stress field near the crack tip approaches a singular behaviour when the degree of nonlocality reduces.

Acknowledgments

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Appendix A

In accordance with Eq. (1), we assume the orthogonal function, $g_N^{p_1 p_2 \dots p_M}(\xi)$, of a relative function, $f(\mathbf{x}')-f(\mathbf{x})$, satisfies the following equation

$$\frac{\partial^{p_1+p_2+\dots+p_M} f(\mathbf{x})}{\partial x_1^{p_1} \partial x_2^{p_2} \dots \partial x_M^{p_M}} = \int_{H_x} [f(\mathbf{x}') - f(\mathbf{x})] g_N^{p_1 p_2 \dots p_M}(\xi) dV_{x'} \quad (\text{A.1})$$

where $p_i \in \mathbf{N}, i=1, 2, \dots, M$ and $1 \leq \sum_{i=1}^M p_i \leq N$. The PD function is constructed as

$$g_N^{p_1 p_2 \dots p_M}(\xi) = \sum_{q_1 + q_2 + \dots + q_M = 1}^N a_{q_1 q_2 \dots q_M}^{p_1 p_2 \dots p_M} w_{q_1 q_2 \dots q_M}(|\xi|) \xi_1^{q_1} \xi_2^{q_2} \dots \xi_M^{q_M} \quad (\text{A.2})$$

Substituting Eqs. (A.1) and (A.2) into Eq. (1) and considering the orthogonality property, one has

$$\sum_{q_1 + q_2 + \dots + q_M = 1}^N A_{(n_1 n_2 \dots n_M)(q_1 q_2 \dots q_M)} a_{q_1 q_2 \dots q_M}^{p_1 p_2 \dots p_M} = b_{n_1 n_2 \dots n_M}^{p_1 p_2 \dots p_M} \quad (\text{A.3})$$

where $A_{(n_1 n_2 \dots n_M)(q_1 q_2 \dots q_M)}$ and $b_{n_1 n_2 \dots n_M}^{p_1 p_2 \dots p_M}$ are same as Eqs. (5a) and (5b), but

$$n_i \in \mathbf{N}, i=1, 2, \dots, M, \quad 1 \leq \sum_{i=1}^M n_i \leq N.$$

Appendix B

The discretized form of Eq. (17) can be written as

$$\begin{aligned} & \sum_{j=1}^{N(k)} \left\{ (u_{1(j)} - u_{1(k)}) \left[(\bar{\lambda}_{(k)} + 2\mu_{(k)}) g_{N(j)(k)}^{20} + \mu_{(k)} g_N^{02} + \left(\frac{d\bar{\lambda}_{(k)}}{dx_1} + 2 \frac{d\mu_{(k)}}{dx_1} \right) g_{N(j)(k)}^{10} \right] + \right. \\ & (u_{2(j)} - u_{2(k)}) \left[(\bar{\lambda}_{(k)} + \mu_{(k)}) g_{N(j)(k)}^{11} + \frac{d\bar{\lambda}_{(k)}}{dx_1} g_{N(j)(k)}^{01} \right] - \\ & \left. \bar{\beta}_{(k)} (T_{(j)} - T_{(k)}) g_{N(j)(k)}^{10} \right\} A - \frac{d\bar{\beta}_{(k)}}{dx_1} (T_{(k)} - T_0) = 0 \\ & \sum_{j=1}^{N(k)} \left\{ (u_{2(j)} - u_{2(k)}) \left[(\bar{\lambda}_{(k)} + 2\mu_{(k)}) g_{N(j)(k)}^{02} + \mu_{(k)} g_{N(j)(k)}^{20} + \frac{d\mu_{(k)}}{dx_1} g_{N(j)(k)}^{10} \right] + \right. \\ & (u_{1(j)} - u_{1(k)}) \left[(\bar{\lambda}_{(k)} + \mu_{(k)}) g_{N(j)(k)}^{11} + \frac{d\mu_{(k)}}{dx_1} g_{N(j)(k)}^{01} \right] - \bar{\beta}_{(k)} (T_{(j)} - T_{(k)}) g_{N(j)(k)}^{01} \left. \right\} A = 0 \\ & \sum_{j=1}^{N(k)} (T_{(j)} - T_{(k)}) \left[k_{(k)} (g_{N(j)(k)}^{20} + g_{N(j)(k)}^{02}) + \frac{dk_{(k)}}{dx_1} g_{N(j)(k)}^{10} \right] A = 0 \end{aligned} \quad (\text{B.1})$$

at $-W/2 \leq x_{1(k)}, x_{1(k)} \leq W/2$

where the subscript (k) and (j) stand for the value of the function at the points $(x_{1(k)}, x_{2(k)})$ and $(x_{1(j)}, x_{2(j)})$, respectively. $N_{(k)}$ denotes as the number of family members of the point $(x_{1(k)}, x_{2(k)})$, and the PD function $g_{N(j)(k)}^{P_1 P_2}$ represents

$$g_{N(j)(k)}^{P_1 P_2} = g_N^{P_1 P_2}(\xi_{1(j)(k)}, \xi_{2(j)(k)}) = g_N^{P_1 P_2}(x_{1(j)} - x_{1(k)}, x_{2(j)} - x_{2(k)}) \quad (\text{B.2})$$

Similarly, the discretized form of Eq. (19) can be expressed as

$$\begin{aligned} \sigma_{11(k)} &= \sum_{j=1}^{N_{(k)}} \left[\left(\bar{\lambda}_{(k)} + 2\mu_{(k)} \right) (u_{1(j)} - u_{1(k)}) g_{N(j)(k)}^{10} + \right. \\ &\quad \left. \bar{\lambda}_{(k)} (u_{2(j)} - u_{2(k)}) g_{N(j)(k)}^{01} \right] A - \bar{\beta}_{(k)} (T_{(k)} - T_0) \\ \sigma_{22(k)} &= \sum_{j=1}^{N_{(k)}} \left[\left(\bar{\lambda}_{(k)} + 2\mu_{(k)} \right) (u_{2(j)} - u_{2(k)}) g_{N(j)(k)}^{01} + \right. \\ &\quad \left. \bar{\lambda}_{(k)} (u_{1(j)} - u_{1(k)}) g_{N(j)(k)}^{10} \right] A - \bar{\beta}_{(k)} (T_{(k)} - T_0) \\ \sigma_{12(k)} &= \sigma_{21(k)} = \mu_{(k)} \sum_{j=1}^{N_{(k)}} \left[(u_{1(j)} - u_{1(k)}) g_{N(j)(k)}^{01} + (u_{2(j)} - u_{2(k)}) g_{N(j)(k)}^{10} \right] A \end{aligned} \quad (\text{B.3})$$

The boundary conditions, Eq. (20), can be discretized as

$$\begin{aligned} \sigma_{22(k)} &= \sigma_{12(k)} = 0, T_{2(k)} = 0, \text{ for } -\frac{W}{2} \leq x_{1(k)} \leq \frac{W}{2}, \text{ and } x_{2(k)} = \pm \frac{W}{2} \\ \sigma_{11(k)} &= -P, \sigma_{12(k)} = 0, T_{(k)} = T_L, \text{ for } x_{1(k)} = -\frac{W}{2}, \text{ and } -\frac{W}{2} \leq x_{2(k)} \leq \frac{W}{2} \\ u_{1(k)} &= \sigma_{12(k)} = 0, T_{(k)} = T_0, \text{ for } x_{1(k)} = \frac{W}{2}, \text{ and } -\frac{W}{2} \leq x_{2(k)} \leq \frac{W}{2} \\ u_{2(k)} &= 0, \text{ for } x_{1(k)} = \frac{W}{2}, \text{ and } x_{2(k)} = 0 \end{aligned} \quad (\text{B.4})$$

Substituting Eq. (B.3) into Eq. (B.4), the discretized form of boundary conditions in terms of displacement components can be obtained. The discretized form of Eq. (5a) can be written as

$$A_{(n_1 n_2)(q_1 q_2)}^{(k)} = \sum_{j=1}^{N_{(k)}} w(|\xi|) \xi_{1(j)(k)}^{n_1 + q_1} \xi_{2(j)(k)}^{n_2 + q_2} A \quad (\text{B.5})$$

where $n_1, n_2 \in \mathbb{N}$ and $1 \leq n_1 + n_2 \leq N$.

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